1. What does the term 'Orthographic' literally signify in technical drawing?

(A) Right-angled drawing (projectors are mutually perpendicular to projection planes)
(B) Inclined drawing
(C) Curved drawing
(D) Perspective drawing

2. In orthographic projection, what is the geometric relationship between projecting rays and the projection plane?

(A) Projecting rays are perpendicular (normal) to the projection plane
(B) Projecting rays are parallel to the plane
(C) Projecting rays are inclined at 45 degrees
(D) Projecting rays converge at a vanishing point

3. Where is the object situated relative to the observer and projection plane in the First Angle Projection system?

(A) Object is placed between the observer and the projection plane
(B) Projection plane is placed between observer and object
(C) Object is behind the observer
(D) Observer is inside the object

4. In First Angle Projection, where is the Top View (Plan) positioned relative to the Front View (Elevation)?

(A) Directly below the front view
(B) Directly above the front view
(C) To the right of the front view
(D) To the left of the front view

5. In First Angle Projection, where is the Left Hand Side View (LHSV) drawn on the drawing sheet?

(A) On the right side of the front view
(B) On the left side of the front view
(C) Above the top view
(D) Below the front view

6. In First Angle Projection, where is the Right Hand Side View (RHSV) located?

(A) On the left side of the front view
(B) On the right side of the front view
(C) Above the front view
(D) Below the front view

7. Where is the projection plane located relative to observer and object in Third Angle Projection?

(A) Projection plane is placed between the observer and the object (transparent plane)
(B) Object is between observer and plane
(C) Observer is between plane and object
(D) Plane is behind infinity

8. In Third Angle Projection, where is the Top View placed relative to the Front View?

(A) Directly above the front view
(B) Directly below the front view
(C) To the right of the front view
(D) To the left of the front view

9. In Third Angle Projection, where is the Left Hand Side View drawn?

(A) On the left side of the front view
(B) On the right side of the front view
(C) Above the front view
(D) Below the top view

10. In Third Angle Projection, where is the Right Hand Side View placed?

(A) On the right side of the front view
(B) On the left side of the front view
(C) Above the top view
(D) Below the front view

11. Which projection system is officially adopted by the Bureau of Indian Standards (BIS) in SP 46:2003?

(A) First Angle Projection System
(B) Third Angle Projection System
(C) Second Angle Projection System
(D) Fourth Angle Projection System

12. Which projection system is standard industrial practice in the United States and Canada (ANSI/ASME Y14.3)?

(A) Third Angle Projection System
(B) First Angle Projection System
(C) Isometric Projection System
(D) Oblique Projection System

13. What geometric solid is used to construct the international graphical symbols for 1st and 3rd angle projection?

(A) Frustum of a right circular cone
(B) Frustum of a square pyramid
(C) Cylinder with keyway
(D) Hemisphere

14. How many principal views can be generated in a standard multi-view orthographic projection box?

(A) 6 principal views
(B) 3 principal views
(C) 4 principal views
(D) 8 principal views

15. Which two primary dimensions are displayed in the Front View (Elevation)?

(A) Width (length) and Height
(B) Width and Depth
(C) Height and Depth
(D) Circumference and Radius

16. Which two primary dimensions are displayed in the Top View (Plan)?

(A) Width (length) and Depth
(B) Width and Height
(C) Height and Depth
(D) Thickness and Angle

17. Which two primary dimensions are displayed in Side Views (End Views)?

(A) Depth and Height
(B) Width and Height
(C) Width and Depth
(D) Area and Volume

18. Which dimension is identical and shared between the Top View and Side Views?

(A) Depth
(B) Height
(C) Width
(D) Weight

19. Which dimension is identical and shared between the Front View and Side Views?

(A) Height
(B) Depth
(C) Width
(D) Perimeter

20. What is the primary criterion for selecting which face of an object to draw as the Front View?

(A) The face that reveals the maximum characteristic shape and minimum hidden lines
(B) The face with the smallest surface area
(C) The bottom face touching the ground
(D) The face that faces Magnetic North

21. Which line type is used to represent hidden edges and invisible contours in orthographic views?

(A) Dashed thin line (Type E or F)
(B) Continuous thick line
(C) Chain thin line
(D) Continuous thin freehand line

22. What line type is used to indicate the axis of symmetry and center of cylindrical holes?

(A) Chain thin line (Long dash-dot, Type G)
(B) Dashed thin line
(C) Continuous thin line
(D) Continuous thick line

23. In manual drafting, what is the standard inclined line used to transfer depth dimensions between Top and Side views?

(A) 45-degree miter line
(B) 30-degree line
(C) 60-degree line
(D) 75-degree line

24. In AutoCAD, which command allows loading and managing linetypes like HIDDEN and CENTER?

(A) LINETYPE (or LTYPE)
(B) LINE
(C) STYLE
(D) LAYER

25. Which system variable in AutoCAD controls the global scaling factor of non-continuous linetypes?

(A) LTSCALE
(B) CELTSCALE
(C) PSLTSCALE
(D) DIMSCALE

26. Which AutoCAD command allows shifting or relocating existing objects from a base point to a destination point?

(A) MOVE (M)
(B) OFFSET
(C) ALIGN
(D) TRANSLATE

27. Which AutoCAD command rotates selected entities about a specified base point through a defined angle?

(A) ROTATE (RO)
(B) SPIN
(C) REVOLVE
(D) TWIST

28. Which AutoCAD command trims objects to meet the cutting edges of other selected entities?

(A) TRIM (TR)
(B) CUT
(C) SPLIT
(D) BREAK

29. Which AutoCAD command extends objects to meet the boundary edges of other entities?

(A) EXTEND (EX)
(B) LENGTHEN
(C) STRETCH
(D) SCALE

30. Which AutoCAD command creates duplicate copies of selected objects at specified vector displacements?

(A) COPY (CO or CP)
(B) CLONE
(C) DUPLICATE
(D) REPLICATE

31. Which AutoCAD command removes unwanted entities permanently from the drawing database?

(A) ERASE (E)
(B) DELETE
(C) PURGE
(D) CLEAR

32. Which AutoCAD command creates a reverse mirrored copy of selected objects across a specified symmetry axis line?

(A) MIRROR (MI)
(B) REFLECT
(C) FLIP
(D) REVERSE

33. Which AutoCAD command enlarges or shrinks selected geometry proportionally about a base point?

(A) SCALE (SC)
(B) ZOOM
(C) RESIZE
(D) STRETCH

34. Which AutoCAD command rounds or smooths the intersection of two lines with an arc of specified radius?

(A) FILLET (F)
(B) CHAMFER
(C) ROUND
(D) BLEND

35. Which AutoCAD command bevels the sharp corner of two intersecting lines with specified setback distances?

(A) CHAMFER (CHA)
(B) FILLET
(C) BEVEL
(D) MITER

36. Which AutoCAD command creates repeated rectangular or circular patterns of objects automatically?

(A) ARRAY (AR)
(B) COPY
(C) PATTERN
(D) OFFSET

37. Which option of the ARRAY command arranges repeated items in rows and columns?

(A) Rectangular Array
(B) Polar Array
(C) Path Array
(D) Linear Array

38. Which option of the ARRAY command arranges repeated items radially around a center point of rotation?

(A) Polar Array
(B) Rectangular Array
(C) Path Array
(D) Spherical Array

39. Which AutoCAD command matches the properties (layer, color, linetype) of a source entity to target entities?

(A) MATCHPROP (MA)
(B) PROPERTIES
(C) CHPROP
(D) COPY

40. In AutoCAD, which mouse selection method (dragged from Left to Right) selects ONLY entities completely enclosed within the rectangle?

(A) Window Selection (Blue Box)
(B) Crossing Selection (Green Box)
(C) Fence Selection
(D) Lasso Selection

41. In AutoCAD, dragging the selection cursor from Right to Left creates a green box with dashed border. This is called:

(A) Crossing Selection (selects all entities touched or enclosed by the boundary)
(B) Window Selection
(C) Fence Selection
(D) Box Selection

42. When a hidden line intersects or joins a visible outline in an orthographic view, how should it be drawn?

(A) The dash of the hidden line should touch the visible line directly without a gap
(B) A 2 mm gap must separate the dash from the visible line
(C) The dash should cross through the visible line
(D) Hidden lines should never meet visible lines

43. When two hidden lines meet to form a corner or T-junction, how should they terminate?

(A) Dashes should meet solidly at the vertex without a gap
(B) A gap should be left at the corner
(C) One dash should cross over the other
(D) They should be connected by a dot

44. When a hidden line runs parallel and adjacent to a visible line, how is clarity maintained?

(A) The dashes of the hidden line should be staggered with a clear gap from the visible line
(B) Both lines should be merged
(C) The hidden line is omitted entirely
(D) The visible line is drawn dashed

45. How should centerlines terminate relative to the circular or symmetrical feature they identify?

(A) They should extend 2 mm to 5 mm past the visible feature outline
(B) They should stop exactly at the perimeter
(C) They should extend to the sheet border
(D) They should stop inside the feature

46. For small circular holes (e.g. diameter < 10 mm), how is the center conventionally drawn on technical sheets?

(A) Drawn as two short intersecting continuous thin line segments (+ cross)
(B) Drawn as long chain dash-dot
(C) Drawn as a solid filled dot
(D) Omitted entirely

47. In AutoCAD, what does typing 'R' inside the ROTATE command activate?

(A) Reference angle option (allows rotating an entity by specifying current angle and target angle)
(B) Relative mode
(C) Radius mode
(D) Reverse direction

48. In AutoCAD, what does typing 'R' inside the SCALE command activate?

(A) Reference scaling option (allows scaling geometry so that a reference length matches a new target length)
(B) Radius scale
(C) Reverse scale
(D) Rectangular scale

49. When using the FILLET command in AutoCAD, what happens if the fillet radius is set to 0 (R = 0)?

(A) Sharpens the corner by extending or trimming the two lines to a precise sharp intersection vertex
(B) Deletes both lines
(C) Throws an error message
(D) Draws a 0 mm circle

50. Which option of the FILLET command applies the rounding radius to ALL vertices of a 2D polyline simultaneously in one click?

(A) Polyline (P) option
(B) Multiple (M) option
(C) Trim (T) option
(D) All option

51. In AutoCAD, which option in the CHAMFER command allows specifying the bevel via one setback distance and an inclination angle?

(A) Angle (A) option
(B) Distance (D) option
(C) Method (M) option
(D) Polyline (P) option

52. What is an 'Associative Array' in AutoCAD?

(A) An array where all elements remain linked as a single parametric object, allowing row/column counts and spacing to be modified later via grips
(B) An array that cannot be edited
(C) An array that links to external Excel sheets
(D) An array that works only in 3D

53. If an associative array in AutoCAD needs to be edited as individual, separate components, which command is executed on it?

(A) EXPLODE (X)
(B) UNLINK
(C) BREAK
(D) SEPARATE

54. In the TRIM command of AutoCAD (Quick Mode), how can a user instantly switch to EXTEND without exiting the command?

(A) Hold down the SHIFT key while clicking entities
(B) Hold down the CTRL key
(C) Hold down the ALT key
(D) Right-click mouse wheel

55. In AutoCAD, what does the 'Fence' (F) selection option inside TRIM or ERASE do?

(A) Draws a multi-segment vector line that trims or selects all entities crossed by the line
(B) Creates a wooden fence block
(C) Encloses objects in a circle
(D) Locks selected entities

56. When creating a Polar Array of 6 holes distributed over a complete circle, what is the fill angle entered?

(A) 360 degrees
(B) 180 degrees
(C) 60 degrees
(D) 90 degrees

57. In the Polar Array of Q56, what is the angular spacing between adjacent holes?

(A) 60 degrees (360 / 6)
(B) 30 degrees
(C) 45 degrees
(D) 90 degrees

58. What does the 'Rotate items' option do in a Polar Array?

(A) Rotates each element as it is arrayed around the center point so it maintains radial orientation
(B) Leaves elements in original upright orientation
(C) Spins each item about its own centroid only
(D) Rotates the entire drawing canvas

59. In AutoCAD, what is the command shortcut for the PROPERTIES palette?

(A) PR (or CTRL+1)
(B) PP
(C) PROP
(D) OP

60. Which command allows quick modification of layer, color, or linetype via command line without opening a palette?

(A) CHPROP
(B) EDIT
(C) MODIFY
(D) ATTEDIT

61. When projecting a cylinder whose axis is perpendicular to HP, what are its orthographic views?

(A) Top view is a true circle; Front view and Side views are rectangles
(B) Top view is a rectangle; Front view is a circle
(C) All three views are circles
(D) All three views are rectangles

62. When projecting a right circular cone with its axis perpendicular to VP, what are its orthographic views?

(A) Front view is a true circle; Top view and Side views are isosceles triangles
(B) Top view is a circle; Front view is a triangle
(C) All views are triangles
(D) Top view is a parabola

63. When projecting a sphere of diameter D, what are its front, top, and side views?

(A) All three orthographic views are identical circles of diameter D
(B) Front view is a circle, top view is an ellipse
(C) All views are ellipses
(D) Front view is a circle, top view is a point

64. When a planar surface of an object is inclined to one projection plane and perpendicular to the other two, its views are:

(A) An inclined edge line in one view, and foreshortened rectangular surfaces in the other two views
(B) True shape in all views
(C) Point views in two planes
(D) Curved arcs in all views

65. What is a 'True Shape' view of an inclined planar surface?

(A) A view projected onto a plane parallel to the surface, displaying all true lengths and true interior angles
(B) A view showing apparent angles
(C) A perspective sketch
(D) An isometric drawing

66. How is a curved surface with a smooth tangent transition to an adjacent flat planar face represented in orthographic projection?

(A) No line is drawn at the tangent intersection (or a thin runout line is shown if the contour blends into a fillet)
(B) A thick solid line is drawn
(C) A dashed hidden line is drawn
(D) A chain thin line is drawn

67. In orthographic drawings, what does a runout curve represent?

(A) The geometric intersection curve formed where a filleted or rounded edge blends into a contoured surface
(B) A manufacturing crack
(C) A dynamic vibration trace
(D) A lead-in for welding

68. In AutoCAD, what command is used to set the current entity creation color without altering the layer setting?

(A) COLOR (though best practice is ByLayer)
(B) RGB
(C) HUE
(D) PALETTE

69. In AutoCAD, what command allows creating multiple parallel offset lines simultaneously at varying distances?

(A) OFFSET (with Layer and Erase source options)
(B) ARRAY
(C) COPY
(D) PARALLEL

70. What does the 'Erase source object after mirroring' prompt in the MIRROR command default to?

(A) [No] (retains the original object while creating the reflected copy)
(B) [Yes]
(C) [Cancel]
(D) [Invert]

71. In AutoCAD, which command allows setting a global scale for dimensions so that text and arrows scale up on large sheets?

(A) DIMSCALE
(B) LTSCALE
(C) TEXTSIZE
(D) SCALE

72. When an object has a cylindrical hole passing entirely through it, how is the hole represented in the view parallel to the hole axis?

(A) Two parallel dashed thin lines representing the hidden hole walls, and a central chain thin line representing the axis
(B) Two solid lines
(C) A shaded gray box
(D) A single centerline only

73. When an object has a blind hole (hole that does not go all the way through), how is the drill point represented?

(A) A conical bottom with an included angle of 118 degrees (drawn at 30 degrees to the horizontal in practice)
(B) A flat flat bottom at 90 degrees
(C) A spherical hemispherical cup
(D) A 45 degree bevel

74. In AutoCAD, what is the shortcut to repeat the last executed command instantly?

(A) Press ENTER or SPACEBAR
(B) Press ESC
(C) Press CTRL+Z
(D) Press TAB

75. What does the command 'OVERKILL' in AutoCAD do?

(A) Removes duplicate or overlapping lines, arcs, and polylines, cleaning up messy imported drawings
(B) Deletes all entities on layer 0
(C) Kills running background scripts
(D) Resets AutoCAD to factory defaults

76. An object has overall dimensions: Length (X) = 120 mm, Depth (Y) = 80 mm, and Height (Z) = 60 mm. If drawn in First Angle Projection with 30 mm spacing between views and 20 mm margins on an A3 sheet (420x297 mm), what is the total horizontal span occupied by the Front View and RHSV?

(A) 230 mm (Front View 120 + Spacing 30 + RHSV Depth 80 = 230 mm)
(B) 200 mm
(C) 260 mm
(D) 180 mm

77. For the object in Q76, what is the total vertical span occupied by the Front View and Top View on the sheet?

(A) 170 mm (Front View Height 60 + Spacing 30 + Top View Depth 80 = 170 mm)
(B) 140 mm
(C) 200 mm
(D) 150 mm

78. In AutoCAD, a rectangular array is specified with 4 rows and 5 columns. Row spacing is 25 mm and column spacing is 40 mm. What is the total bounding box size from bottom-left element center to top-right element center?

(A) 160 mm along X and 75 mm along Y ((5-1)*40 = 160 mm; (4-1)*25 = 75 mm)
(B) 200 mm along X and 100 mm along Y
(C) 160 mm along X and 100 mm along Y
(D) 120 mm along X and 50 mm along Y

79. In AutoCAD, a polar array of 8 equally spaced bolt holes is created on a Pitch Circle Diameter (PCD) of 150 mm. What is the straight-line chord distance between adjacent hole centers?

(A) 57.40 mm (2 * 75 * sin(360? / (2 * 8)) = 150 * sin(22.5?) = 150 * 0.3827 = 57.40 mm)
(B) 75.00 mm
(C) 50.00 mm
(D) 62.83 mm

80. A line segment of length 73.48 mm needs to be scaled up in AutoCAD so that its length becomes exactly 120.00 mm. Using the SCALE command with the Reference option, what are the reference and new length inputs?

(A) Reference length = 73.48; New length = 120.00
(B) Scale factor = 120 / 73.48 = 1.6331
(C) Reference length = 120; New length = 73.48
(D) Both A and B are mathematically valid workflows in AutoCAD

81. In AutoCAD, to rotate a line segment currently inclined at 37 degrees to the horizontal so that it becomes inclined at exactly 90 degrees (vertical), what is the input in ROTATE -> Reference?

(A) Specify reference angle: 37; Specify new angle: 90
(B) Specify angle: 53
(C) Specify reference angle: 90; Specify new angle: 37
(D) Both A and B achieve the identical orientation

82. A truncated square pyramid (base 50 mm, top 25 mm, height 60 mm) has its axis perpendicular to HP and base edges equally inclined at 45 degrees to VP. What is the shape of its top view?

(A) Two concentric squares with sides 50 mm and 25 mm whose edges are inclined at 45 degrees to XY, with corner diagonals connecting them
(B) A single square
(C) Two concentric octagons
(D) A trapezoid

83. For the pyramid in Q82, what is the appearance of the Front View?

(A) A symmetrical trapezoid with a central vertical line representing the nearest corner edge
(B) A flat rectangle
(C) A triangle
(D) Two overlapping squares

84. In orthographic multi-view generation, when an inclined circular hole is projected onto a plane not parallel to its circular face, what geometric curve is formed?

(A) An ellipse (whose minor axis equals diameter * cos(tilt angle) and major axis equals true diameter)
(B) A smaller true circle
(C) A parabola
(D) An oval with straight sides

85. A horizontal cylinder of diameter 40 mm penetrates through a vertical square prism of side 50 mm such that their axes intersect at right angles. What is the curve of intersection seen in the front view?

(A) Two symmetric circular arcs (curves of intersection)
(B) A straight vertical line
(C) A true ellipse
(D) A rectangular box

86. In AutoCAD, which command creates a 2D planar Region from a closed boundary of lines, enabling subsequent Boolean Union, Subtract, and mass property analysis?

(A) REGION (REG)
(B) BOUNDARY
(C) GROUP
(D) BLOCK

87. What is the primary difference between a REGION and a closed POLYLINE in AutoCAD?

(A) A Region is an actual 2D planar solid surface that supports Boolean operations (Union/Subtract), while a polyline is a wireframe boundary
(B) Polylines have area while regions do not
(C) Regions cannot have curved edges
(D) Regions work only in 3D perspective

88. In AutoCAD, to trim multiple lines against a cutting edge by drawing a freehand cutting path through them, which sub-option of TRIM is used?

(A) Fence (F) or Crossing (C)
(B) Project
(C) Edge
(D) Extend

89. A stepped shaft has diameters 50 mm and 30 mm joined by a 5 mm fillet radius. In orthographic front view, how is this transition drafted?

(A) A continuous circular arc of radius 5 mm tangent to the vertical shoulder face and horizontal 30 mm shaft surface
(B) A 45 degree chamfer line
(C) A dashed line
(D) A vertical line

90. In First Angle Projection, if an object has a cylindrical counterbored hole from the top face, how does the counterbore appear in the Front View?

(A) As two steps of parallel dashed lines: wider dashed lines for the counterbore depth, and narrower dashed lines extending through the remainder
(B) As two solid steps
(C) As a single dashed line
(D) As a hatched region

91. In AutoCAD, what does the command 'JOIN' (J) do when executed on two collinear, separated line segments?

(A) Merges them into a single continuous straight line segment spanning the gap
(B) Draws a perpendicular connecting line
(C) Creates a block
(D) Converts them to an arc

92. In AutoCAD, which command breaks a single object at a specific picked point without leaving an empty gap?

(A) BREAK -> select object -> type 'F' (First point) -> pick point -> type '@' (at current point)
(B) TRIM
(C) SPLIT
(D) EXPLODE

93. In AutoCAD, what happens when you use 'CHAMFER' with Distance 1 = 10 mm and Distance 2 = 15 mm?

(A) Creates an asymmetrical bevel cutting 10 mm back along the first selected line and 15 mm back along the second selected line
(B) Creates a symmetrical 12.5 mm bevel
(C) Creates a 45 degree cut
(D) Rejects the command

94. When an inclined planar face of an object makes an angle of 60 degrees with the horizontal plane (HP), what is the foreshortening factor for its depth in the Top View?

(A) cos(60?) = 0.50 (depth appears at 50% of its true physical dimension)
(B) sin(60?) = 0.866
(C) tan(60?) = 1.732
(D) 1.0 (no foreshortening)

95. In orthographic drafting, what is the conventional representation of external screw threads on a shaft in front view?

(A) Crests drawn as continuous thick lines, roots drawn as continuous thin lines (Type B) spaced at nominal pitch
(B) Both crests and roots drawn thick
(C) Threads drawn as zigzag sawteeth
(D) Threads shown as hidden dashed lines

96. In orthographic drafting, how is an internal threaded hole (tapped hole) represented in elevation when looking perpendicular to its axis?

(A) Bore diameter shown as dashed thick lines, thread root diameter shown as parallel dashed thin lines
(B) Thread root shown thick
(C) Both shown solid
(D) Only centerline is shown

97. In AutoCAD, how can you temporarily toggle Ortho mode while drawing without clicking the F8 key or status bar button?

(A) Hold down the SHIFT key while moving the mouse cursor
(B) Hold down CTRL
(C) Hold down ALT
(D) Press mouse scroll wheel

98. What does the command 'LAYISO' (Layer Isolate) do in an AutoCAD drafting session?

(A) Hides or locks all layers EXCEPT the layers of the selected objects, allowing focused editing without background clutter
(B) Deletes all unselected layers
(C) Turns off layer 0
(D) Exports selected layers to DXF

99. What command restores all layers turned off by LAYISO?

(A) LAYUNISO
(B) LAYER ALL
(C) LAYON
(D) RESTORE

100. Why is Third Angle Projection universally favored in North American robotics and computer numerical control (CNC) machining?

(A) Because the arrangement of orthographic views corresponds directly to the intuitive physical layout of machining axes (X-Y table movements and natural line of sight)
(B) Because it uses fewer drawing sheets
(C) Because it eliminates the need for dimensions
(D) Because it is required by English common law

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the fundamental principles of orthographic projection, detailing how a three-dimensional object is mapped onto two-dimensional planes using parallel normal projecting rays.

Q2. Compare and contrast First Angle Projection and Third Angle Projection systems in exhaustive detail, including relative positions of views, international symbols, and regional adoption.

Q3. Detail the six standard principal views of an object in orthographic projection, explaining their geometric relationships and the dimensional alignment between them.

Q4. Explain the methods of transferring depth dimensions between the Top View and Side Views in manual and CAD drafting: the 45-degree Miter Line Method and the Compass Arc Method.

Q5. Discuss the rules and conventions for representing hidden features (dashed lines) in orthographic drawings according to BIS SP 46 and ISO 128, including intersection etiquette.

Q6. Describe the conventions for representing cylindrical features, circular holes, and pitch circle diameters in orthographic views, detailing centerlines and cross marks.

Q7. Explain the selection criteria for the Front View (Elevation) of an engineering component, detailing how orientation impacts drawing clarity and hidden line density.

Q8. Explain the LINETYPE and LTSCALE commands in AutoCAD, discussing how linetype files (.lin) are structured, loaded, and scaled for model space and paper space layouts.

Q9. Detail the MOVE and COPY commands in AutoCAD, contrasting displacement vector entry with base-point pick methods and discussing the Array sub-option of COPY.

Q10. Detail the ROTATE command in AutoCAD, explaining base-point selection, positive/negative angular conventions, the Copy sub-option, and the Reference angle option.

Q11. Detail the TRIM and EXTEND commands in AutoCAD, explaining Cutting/Boundary edge selection, Standard vs Quick mode, and the Shift-key toggle behavior.

Q12. Detail the FILLET and CHAMFER commands in AutoCAD, comparing their geometric functions, polyline application, trim/no-trim modes, and zero-radius corner cleanup.

Q13. Explain the ARRAY command in AutoCAD, providing comprehensive mathematical and operational details for Rectangular Array, Polar Array, and Path Array.

Q14. Detail the MIRROR and SCALE commands in AutoCAD, explaining symmetry axis definition, source object retention, base point selection, and proportional resizing.

Q15. Explain the MATCHPROP command and the PROPERTIES palette in AutoCAD, detailing how object attributes (layer, linetype, color, lineweight) are inspected and transferred.

Q16. A cast iron bearing bracket has a base 120 mm x 60 mm x 20 mm thick, a central vertical upright 60 mm wide x 20 mm thick x 50 mm high, and a horizontal cylindrical hole of 30 mm diameter drilled centrally through the upright. Provide a detailed step-by-step description of its Front View, Top

Q17. Explain the procedure to convert an isometric pictorial drawing of a stepped machine block with a V-groove and countersunk hole into complete orthographic views in AutoCAD.

Q18. Discuss the representation of intersections of curved surfaces in orthographic projection, detailing the line of intersection between two intersecting cylinders of equal and unequal diameters.

Q19. Detail the selection set modes in AutoCAD: Window, Crossing, Fence, Window Polygon (WP), Crossing Polygon (CP), and Previous (P), with specific drafting scenarios for each.

Q20. Explain the methods of creating and editing rectangular arrays in AutoCAD using both the command-line interface and the contextual Ribbon Array tab.

Q21. A stepped shaft has three concentric cylindrical sections: Section 1 (diameter 60 mm, length 40 mm), Section 2 (diameter 40 mm, length 60 mm), and Section 3 (diameter 25 mm, length 50 mm). Section 2 has a central keyway 10 mm wide x 4 mm deep x 30 mm long. Describe its Front View and

Q22. Explain the difference between First Angle and Third Angle projection symbols, providing exact geometric proportions for drawing them according to BIS SP 46.

Q23. Explain the methods of drawing runouts and fillet blend transitions in orthographic views, detailing when lines are drawn and when they are omitted.

Q24. Detail the STRETCH command in AutoCAD, explaining crossing-window selection rules, vertex translation, and how to resize multi-view orthographic features consistently.

Q25. Explain the procedure to dimension orthographic multi-view drawings in AutoCAD according to ISO standards, detailing baseline vs continuous dimensioning.

Q26. A hollow hexagonal prism (base side 40 mm, height 80 mm, central circular hole of diameter 35 mm through its length) rests on its base on HP with one base edge parallel to VP. Describe its orthographic Front View, Top View, and Left Hand Side View in First Angle projection.

Q27. Explain the methods of creating and managing AutoCAD Blocks (BLOCK, BMAKE, WBLOCK, INSERT) for standard orthographic symbols and hardware components.

Q28. Explain the use of construction lines (XLINE and RAY) in AutoCAD for establishing multi-view projection alignments, detailing their infinite geometry and sub-options.

Q29. Explain the procedure to create an aligned Auxiliary View for an object having an inclined face, detailing why principal views fail to show the true shape of inclined surfaces.

Q30. Summarize the complete engineering quality audit checklist for validating orthographic multi-view drawings prior to manufacturing release.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[D]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[D]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- Adoption: Mandatory standard across Europe, India (BIS SP 46:2003), ISO member nations, and Commonwealth countries.

2. Third Angle Projection System:
- Principle: The object is placed in the Third Quadrant. The transparent projection plane is situated directly BETWEEN the observer and the object.
- Projection Mechanics: The observer views the object through the transparent plane; the image is traced directly onto the surface facing the observer.
- Arrangement of Views on Sheet:
* Front View (Elevation) is the central reference.
* Top View (Plan) is drawn ABOVE the front view.
* Left Hand Side View is drawn on the LEFT of the front view.
* Right Hand Side View is drawn on the RIGHT of the front view.
* Bottom View is drawn BELOW the front view.
- Graphical Symbol: Frustum of a cone with the two concentric circles positioned on the LEFT, and the trapezoidal profile on the RIGHT.
- Adoption: Official industrial standard in USA and Canada (ANSI/ASME Y14.3) and widely used in automotive CAD.

>> **Solution to Question 3:** Detail the six standard principal views of an object in orthographic projection, explaining their geometric relationships and the dimensio

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- Rule 3 (Hidden T-Junctions):
When a hidden line intersects another hidden line forming a T, the cross dash must touch the stem dash solidly.
- Rule 4 (Crossing Hidden Lines):
When two non-intersecting hidden lines cross in projection (one in front of the other in space), one dash must bridge the gap of the other to prevent the appearance of a false 4-way intersection.
- Rule 5 (Parallel Adjacent Lines):
When a hidden line runs parallel and close to a visible line, stagger the dashes to prevent the appearance of a single blurred line.

3. Omission Rule:
Hidden lines should be omitted when they do not contribute to clarity or when a sectional view is provided that shows those features as visible outlines.

- Criterion 1 (Maximum Shape Characteristic):
Select the viewing direction that reveals the most distinctive profile, primary contours, steps, tapers, and characteristic features of the part.
- Criterion 2 (Minimum Hidden Lines):
Orient the object such that the internal cavities, holes, and slots are visible from the front or situated such that the resulting views contain the minimum possible number of hidden dashed lines.
- Criterion 3 (Natural Operating Orientation):
Whenever practical, orient the object in the position it normally occupies in service or manufacturing (e.g., motor shafts horizontal, engine pistons vertical, brackets seated on their mounting bases).
- Criterion 4 (Longest Dimension Horizontal):
Position the longest linear dimension of the component horizontally. This utilizes the wide landscape format of standard engineering drawing sheets (A4, A3, A2) most efficiently.

2. Practical Impact:
A poorly chosen front view results in cluttered top and side views dominated by overlapping dashed lines, increasing drafting time, visual fatigue, and the likelihood of costly machining errors in the workshop.

[Detailed Step-by-Step Answer]:
1. MOVE Command (M):
- Function: Translates selected entities from one location to another without altering orientation or geometry.
- Command Sequence:
* Command: MOVE [Enter]
* Select objects: Use Window, Crossing, or click individual entities -> [Enter].
* 'Specify base point or [Displacement] <Displacement>:':
- Method A (Base Point Pick): Snap to an existing vertex (e.g., endpoint of line) using OSNAP. Move cursor to target vertex and click.
- Method B (Direct Distance / Coordinate Displacement): Type relative coordinates (e.g. '@50,0' or '@100<45') or turn on Ortho (F8), drag mouse cursor, and enter numerical distance.

2. COPY Command (CO / CP):
- Function: Creates duplicate instances of selected objects while preserving the original entities.
- Command Sequence:
* Command: COPY [Enter] -> Select objects -> [Enter].
* Specify base point -> Specify second point of displacement.
* Default Behavior: Multiple copy mode is active by default; the user can continue clicking second points indefinitely to paste multiple instances until pressing Enter or Esc.

3. The Array Sub-Option in COPY:
- Inside the COPY command, after picking the base point, type 'A' (Array) [Enter].
- Prompts: 'Enter number of items to array:' -> Type e.g. 5 [Enter].
- Prompts: 'Specify second point or [Fit]:'
* Moving cursor and typing distance '30' places 5 copies spaced 30 mm apart along the vector.
* Typing 'F' (Fit) specifies the total overall distance over which all 5 items are spaced equally!

>> Solution to Question 18: Detail the ROTATE command in AutoCAD, explaining base-point selection, positive/negative angular conventions, the Copy sub-option

- Invaluable when an entity is inclined at an unknown arbitrary angle and needs to be aligned to a known target angle (e.g. 0? horizontal or 90? vertical).
- Typing 'R' [Enter] prompts:
* 'Specify the reference angle <0>:': Click two points along the current inclined edge of the object to capture its existing angle.
* 'Specify the new angle or [Points] <0>:': Enter the desired final angle (e.g., '0' to align perfectly horizontal) or click two points on another reference object.
- Eliminates all trigonometric angle calculation and manual error!

>>Solution Question 12: the TRIM and EXTEND commands in AutoCAD, along with the Boundary selection, Standard, Quick, and the

- Polyline Option (P): Rounds ALL interior and exterior corners of a 2D polyline in a single click.

2. CHAMFER Command (CHA):
- Function: Bevels the intersection corner of two lines with a straight sloping transition line.
- Method Options:
* Distance Method (D): Specify First Chamfer Distance (d1) and Second Chamfer Distance (d2).
* Angle Method (A): Specify First Chamfer Distance (d) and Chamfer Angle (theta) relative to first line (e.g. 2 mm x 45?).
- Select first line -> Select second line.

3. Trim Mode (T):
- Both commands support Trim / No Trim mode:
* Trim Mode (Default): The original sharp corner lines are automatically trimmed back to the blend endpoints.
* No Trim Mode: Inserts the fillet arc or chamfer line while preserving the original sharp corner lines intact.

>>Solution to Question 13: Explain the ARRAY command in AutoCAD, providing comprehensive mathematical and operational details for Rectangular Array, Polar

* Measure: Places items at exact specified arc-length intervals along the path.

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applies them to one or more destination objects.
- Command Sequence:
* Command: MATCHPROP [Enter]
* 'Select source object:' -> Click entity with desired properties (e.g., a centerline on the 'CENTER' layer).
* 'Select destination object(s) or [Settings]:' -> Click or window target entities; they instantly adopt the source properties.
- Settings Option (S):
* Opens Property Settings dialog allowing the user to selectively enable or disable matching for Color, Layer, Linetype, Linetype Scale, Lineweight, Thickness, Text Style, Dimension Style, Hatch, and Polyline Width.

>>Solution to Question 18: A cast iron bearing bracket has a base 120 mm x 60 mm x 20 mm thick, a central vertical upright 60 mm wide x 20 mm thick x 50 mm h

[Problem Statement]: Explain the procedure to convert an isometric pictorial drawing of a stepped machine block with a V-groove and countersur
[Detailed Step-by-Step Answer]:
AutoCAD Drafting Sequence:
1. Environment Setup:
- Units: Decimal, Millimeters. Limits: 0,0 to 420,297 (A3). ZOOM ALL.
- Create Layers: 'VISIBLE' (0.5 mm), 'HIDDEN' (0.25 mm, HIDDEN), 'CENTER' (0.25 mm, CENTER), 'CONSTRUCT' (0.18 mm, Continuous), 'DIM' (0.25 mm).

2. Draft Front View (Anchor View):
- Switch to 'VISIBLE' layer.
- Command: LINE -> Draw base line: 0,150 to @100<0.
- Draw stepped vertical outlines, top horizontal plateau, and sloping V-groove boundaries using relative polar entries (e.g. @25<-45, @25<45).
- Draw through-hole and countersunk boundaries as dashed hidden lines on layer 'HIDDEN':
* Command: LINE -> Specify start point -> Draw hidden vertical walls of hole.
* Draw 90? countersunk bevel lines at top surface.
- Add chain thin centerline on layer 'CENTER'.

3. Project Top View:
- Switch to 'CONSTRUCT' layer. Turn on Ortho (F8).
- Command: LINE -> Project vertical construction rays downwards from every vertex, corner, and center in the Front View.
- Switch to 'VISIBLE' layer. Draw top view horizontal boundary lines at appropriate depth offset (e.g. Y = 50 to Y = 110).
- Trim intersection lines using TRIM (TR).
- Where the V-groove passes across the block, draw visible horizontal boundary edges.
- For countersunk hole: Draw two concentric circles using CIRCLE (Center, Radius) on layer 'VISIBLE':
* Inner circle = Through-hole diameter.
* Outer circle = Countersink outer diameter. Add cross centerlines.

4. Construct Side View via 45? Miter Line:
- On 'CONSTRUCT' layer, draw horizontal line from bottom of Front View and vertical line from right of Top View intersecting at origin O.
- Command: LINE -> From O, draw miter ray: @150<45.
- Project all depth features from Top View horizontally to miter line, then vertically upward into Side View.
- Project height features horizontally from Front View to intersect vertical projectors.
- Connect visible outlines on 'VISIBLE' layer; convert internal obscured cavities to 'HIDDEN' layer.
- Turn off or freeze 'CONSTRUCT' layer for clean presentation.

>>**Solution to Question 18:** [Problem Statement]: Discuss the representation of intersections of curved surfaces in orthographic projection, detailing the line of intersection between two i

* In the Front View, the intersection forms a smooth 3D spatial curve (hyperbolic curve) convex towards the axis of the larger cylinder.
- Plotting Method (Cutting Plane Method):
* Pass multiple horizontal cutting planes through both cylinders.
* Each horizontal cutting plane cuts two straight line generators from the small cylinder and a circular arc from the large cylinder.
* The intersection of these generators with the circle gives discrete points on the intersection curve.
* Plot these points in the Front View and connect with a smooth curve using a French Curve or AutoCAD SPLINE.

2. Intersection of Two Cylinders of EQUAL Diameters (d = D):
- When two right circular cylinders of identical diameter intersect with their axes at 90 degrees, the 3D intersection curve degenerates into two planar intersecting ellipses!
- In the Front View, these two intersecting ellipses project as two STRAIGHT LINES intersecting at the center like an 'X'!
- This is an extremely common drafting convention in piping tees and manifold junctions.

Solution to Question 18: the selection set modes in AutoCAD: Window, Crossing, Fence, Window Polygon (WP), Crossing Polygon (CP), and Previous (P

3. End View (Looking from the Right at Section 3, drawn on the LEFT in 1st Angle):
- Centerlines: Mutually perpendicular horizontal and vertical chain thin lines intersecting at the shaft center (+).
- Section 3: An innermost visible circle of diameter 25 mm.
- Section 2: A concentric visible circle of diameter 40 mm.
- Section 1: An outermost concentric visible circle of diameter 60 mm.
- Keyway: Visible in Section 2 at top! A rectangular slot 10 mm wide (5 mm left and right of vertical centerline) extending 4 mm downward from the 40 mm diameter perimeter.

>>Solution to Question 23 Explain the difference between First Angle and Third Angle projection symbols in views, detailing when each is used and drawn the

2. Conventional Representation Rules:
- Rule 1 (Omission of Line):
Where a flat planar surface blends tangentially into a cylinder, cone, or fillet curve, NO LINE is drawn on the orthographic view representing that transition.
- Rule 2 (Runout Curves):
Where a filleted corner intersects or terminates at a curved contoured body (e.g., a web or rib joining a cylindrical boss), the intersection line fades out.
Conventionally, this transition is represented by a partial circular arc or curve called a 'Runout'.
The runout curve terminates tangentially before reaching the main outer boundary.
- Rule 3 (Fictitious Lines of Intersection):
For clarity on complex castings, draftsmen historically used thin continuous lines (~0.25 mm) stopping slightly short of outer contours to delineate major structural flanges that have rounded corners.

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perimeter.
- Feature Association: Dimensions should be attached to the view that displays the true characteristic profile of the feature (e.g. dimension hole diameter on the circular view, not on the dashed rectangular view).
- Spacing: Distance from object outline to first dimension line must be >= 10 mm; spacing between parallel dimension lines must be >= 8 mm.

2. Baseline Dimensioning (DIMBASELINE - DBL):
- Function: Creates a series of parallel dimensions originating from a single common reference datum edge.
- Advantage: Eliminates cumulative manufacturing tolerances; each feature location is machined independently from the primary datum surface.
- Command Sequence: Place initial dimension via DIMLINEAR -> Execute DIMBASELINE -> Click subsequent feature edges. AutoCAD offsets each dimension line by the preset baseline spacing automatically.

3. Continuous (Chain) Dimensioning (DIMCONTINUE - DCO):
- Function: Places dimensions in a continuous end-to-end series along a single straight line.
- Application: Used where the overall length is non-critical, or where step-to-step spacing is functionally paramount.
- Precaution: Never dimension every segment in a closed chain and also provide the overall dimension; omit one non-critical segment to prevent tolerance conflicts.

>>**Solution to Question 26:** A hollow hexagonal prism (base side 40 mm, height 80 mm, central circular hole of diameter 35 mm through its length) rests on its base

- Outer Boundary: A rectangle 69.28 mm wide x 80 mm high.
- Centerline: Vertical chain thin line along axis of symmetry.
- Vertical Edges: Two vertical faces visible; outer edges are flat surfaces.
- Central Hole: Two vertical Dashed Thin Lines spaced 35 mm apart extending through the 80 mm height.

>>Solution to Question 28 Explain the methods of creating lines (XLINE and RAY) and polylines (PLINE) in AutoCAD. Discuss their applications and how they differ from standard lines.

to an existing entity.
* Bisect (B): Bisects the angle formed by three picked points.
* Offset (O): Creates an infinite line parallel to an existing entity at a specified distance.

3. RAY Command:
- Function: Creates a semi-infinite line originating at a picked start point and extending infinitely in one specified direction.
- Application: Projecting depth from Top View to miter line without cluttering upper quadrants.

4. Layer Isolation Best Practice:
- Place all Xlines and Rays on a dedicated layer named 'CONSTRUCT' or 'DEFPOINTS' and toggle layer visibility OFF before final plotting.

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visible outlines at corners and gaps are clean.
- Confirm centerlines extend 2-5 mm past circular features and intersect at long dashes (+).
- Check global LTSCALE and run REGEN to ensure no dashed lines display solid.

3. Dimensioning Integrity:
- Check that every feature is dimensioned exactly once (no redundant over-dimensioning).
- Ensure dimensions are outside views wherever possible.
- Verify diameter symbol (phi) precedes all cylindrical callouts and radius symbol (R) precedes fillet arcs.
- Check that dimension extension lines do not cross dimension lines.

4. Title Block & Regulatory Compliance:
- Verify institutional name, drawing title, drawing number, and revision index.
- Confirm the correct international projection symbol (1st or 3rd angle) is displayed.
- Verify stated drawing scale matches model viewport zoom factor (e.g. 1:1, 1:2).
